Milestone : Implementation in NoSQL

IE 6700 Data Management for Analytics

Group 4
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Submission Date : November 30, 2025

Introduction

This report presents the NoSQL implementation of a Smart Food Delivery & Marketplace Management System using MongoDB Atlas.
The objective of this implementation is to recreate the earlier SQL-based system in a document-oriented NoSQL model, demonstrating flexible schema design, nested documents, embedded arrays, and powerful analytical capabilities through MongoDB Aggregation Pipelines.

The implementation includes:

- Creating a database in MongoDB Atlas
- Building four collections (Customers, Restaurants, Drivers, Orders)
- Inserting realistic sample documents (5–10 per collection)
- Running 5 aggregation pipeline queries for analytics
- Interpreting the results

The system includes creating a MongoDB cluster, building collections, inserting realistic sample documents, and executing advanced queries to extract insights about customers, restaurants, drivers, orders, and payments.

Collection Design and Data Modeling

1. customers Collection - Stores customer information.

Fields Included:

- customer_id
- name
- gender
- date_of_birth
- phone_number
- address
- account_type

Description:
A simple flat structure captures essential customer details for identification and personalization.

2. restaurants Collection - Stores details of restaurants on the platform.

Fields Included:

- restaurant_id
- name
- cuisine_type
- location
- rating

Description:
Each restaurant has its own entry with independent attributes.

3. drivers Collection - Represents delivery personnel on the platform.

Fields Included:

- driver_id
- name
- rating
- contact
- status

Description:
A small collection used for mapping order deliveries to drivers.

4. orders Collection - Orders contain multiple internal components and demonstrate NoSQL's flexible embedding capabilities.

Fields Included:

- order_id
- customer_id
- restaurant_id
- promo_code
- order_time
- total_amount
- status
- items (embedded array)
- payment (embedded object)
- delivery (embedded object)

QUERY-1 Top customers by total spending

```
[
  {
    $group: {
      _id: "$customer_id",
      total_spent: { $sum: "$total_amount" }
    }
  },
  {
    $lookup: {
      from: "customers",
      localField: "_id",
      foreignField: "customer_id",
      as: "customer"
    }
  },
  { $unwind: "$customer" },
  {
    $project: {
      _id: 0,
      customer_id: "$customer.customer_id",
      name: "$customer.name",
      total_spent: 1
    }
  },
  { $sort: { total_spent: -1 } }
]
```

```
total_spent : 27.98
customer_id : 1
name : "Dev Patel"

total_spent : 22.48
customer_id : 3
name : "Ava Brown"

total_spent : 18.98
customer_id : 4
name : "Liam Johnson"

name : "Hitaxi Lethwala"
total_spent : 16.48
customer_id : 2

total_spent : 11.99
customer_id : 5
name : "Noah Davis"
```

Purpose:

To identify which customers generate the most revenue.

Key Actions:

- Group orders by customer
- Sum the total spending
- Lookup customer names
- Sort in descending order

Insight Example:

Premium customer **Dev Patel** spent the highest, showing strong customer value.

QUERY-2 Most Frequently Ordered Items

```
[
  { $unwind: "$items" },
  {
    $group: {
      _id: { item_id: "$items.item_id", item_name: "$items.name" },
      total_quantity: { $sum: "$items.quantity" },
      order_count: { $sum: 1 }
    }
  },
  {
    $project: {
      _id: 0,
      item_id: "$_id.item_id",
      item_name: "$_id.item_name",
      total_quantity: 1,
      order_count: 1
    }
  },
  { $sort: { total_quantity: -1 } }
]
```

```
total_quantity : 1
order_count : 1
item_id : 7
item_name : "California Roll"

item_id : 10
item_name : "Avocado Wrap"
total_quantity : 1
order_count : 1

total_quantity : 1
order_count : 1
item_id : 6
item_name : "Garlic Bread"

total_quantity : 1
order_count : 1
item_id : 1
item_name : "Butter Chicken"

item_name : "Salmon Nigiri"
total_quantity : 1
order_count : 1
item_id : 8

item_name : "Margherita Pizza"
total_quantity : 1
order_count : 1
item_id : 4

total_quantity : 1
order_count : 1
item_id : 2
item_name : "Paneer Tikka"

total_quantity : 1
order_count : 1
item_id : 3
item_name : "Veg Biryani"

item_id : 9
item_name : "Quinoa Salad"
total_quantity : 1
order_count : 1
```

Purpose:

To analyze item popularity.

Key Actions:

- Unwind items array
- Count frequency & quantity
- Sort by quantities

Insight Example:

Fast-moving items help restaurants plan menu and inventory.

QUERY-3 Average Order Value Per Restaurant

```
[
  {
    $group: {
      _id: "$restaurant_id",
      avg_order_value: { $avg: "$total_amount" },
      total_orders: { $sum: 1 }
    }
  },
  {
    $lookup: {
      from: "restaurants",
      localField: "_id",
      foreignField: "restaurant_id",
      as: "restaurant"
    }
  },
  { $unwind: "$restaurant" },
  {
    $project: {
      _id: 0,
      restaurant_id: "$restaurant.restaurant_id",
      restaurant_name: "$restaurant.name",
      avg_order_value: 1,
      total_orders: 1
    }
  },
  { $sort: { avg_order_value: -1 } }
]
```

```
avg_order_value : 22.48
total_orders : 1
restaurant_id : 3
restaurant_name : "Sushi Sensei"

avg_order_value : 19.985
total_orders : 2
restaurant_id : 1
restaurant_name : "Spice Route"

avg_order_value : 18.98
total_orders : 1
restaurant_id : 4
restaurant_name : "Green Bowl"

avg_order_value : 16.48
total_orders : 1
restaurant_id : 2
restaurant_name : "Pizza Plaza"
```

Purpose:

To find which restaurant generates the highest average bill.

Key Actions:

- Group orders by restaurant
- Calculate average
- Lookup restaurant details

Insight Example:

“Sushi Sensei” or “Spice Route” may show higher average because of premium cuisine.

QUERY-4 Deliveries Completed by Each Driver

```
[
  {
    $match: { "delivery.status": "DELIVERED" }
  },
  {
    $group: {
      _id: "$delivery.driver_id",
      deliveries_completed: { $sum: 1 }
    }
  },
  {
    $lookup: {
      from: "drivers",
      localField: "_id",
      foreignField: "driver_id",
      as: "driver"
    }
  },
  { $unwind: "$driver" },
  {
    $project: {
      _id: 0,
      driver_id: "$driver.driver_id",
      driver_name: "$driver.name",
      deliveries_completed: 1
    }
  },
  { $sort: { deliveries_completed: -1 } }
]
```

```
deliveries_completed : 2
driver_id : 1
driver_name : "Raj Mehta"
```

```
driver_name : "Kim Lee"
deliveries_completed : 1
driver_id : 2
```

```
deliveries_completed : 1
driver_id : 3
driver_name : "Carlos Ruiz"
```

```
deliveries_completed : 1
driver_id : 4
driver_name : "Mina Park"
```

Purpose:

Shows driver efficiency and workload.

Key Actions:

- Match delivered orders
- Group by driver
- Lookup driver name
- Sort by completed deliveries

Insight Example:

Driver Raj Mehta completed more deliveries, showing strong performance.

QUERY-5 Payment Method Usage Summary

```
[
  {
    $group: {
      _id: "$payment.method",
      number_of_payments: { $sum: 1 },
      total_amount_processed: { $sum: "$payment.amount" }
    }
  },
  {
    $project: {
      _id: 0,
      payment_method: "$_id",
      number_of_payments: 1,
      total_amount_processed: 1
    }
  },
  { $sort: { total_amount_processed: -1 } }
]
```

```
number_of_payments : 3
total_amount_processed : 63.44
payment_method : "CARD"
```

```
number_of_payments : 1
total_amount_processed : 22.48
payment_method : "WALLET"
```

```
number_of_payments : 1
total_amount_processed : 11.99
payment_method : "CASH_ON_DELIVERY"
```

Purpose:

To summarize payment trends.

Key Actions:

- Group by payment method
- Count total usage
- Sum total revenue from each method

Insight Example:

CARD might be most used, while CASH_ON_DELIVERY could be lower

Conclusion

This NoSQL implementation successfully demonstrates how MongoDB can model a real-world food delivery platform using a flexible, document-oriented data model.
The use of embedded arrays and nested documents within the **orders** collection reduced the need for joins and improved data retrieval performance.
Six aggregation queries were executed to analyze customer spending, item popularity, restaurant performance, driver efficiency, and payment behavior.
Overall, the project validates the advantages of NoSQL systems in analytics, scalability, and schema flexibility.